DATASCI 530 – Assignment 4

For this assignment you will practice the use of regular expressions to extract information from text data. You will be working be real text from bills and actions from the 116th Congress of the United States:

https://en.wikipedia.org/wiki/116th_United_States_Congress

For this homework, you will be working with the "bills_actions.csv" dataset in Lecture 10:

https://github.com/alejandrosanchezbecerra/qtm531spring2024/blob/main/Lecture%2010/data_raw/bills_actions.csv

You will be asked to process unstructured text data and extract specific elements of text.

- Give some basic descriptive statistics from the dataset. How many distinct "bill" identifiers are there? What proportion of the rows belong to different categories? Within each category, calculate the proportion of rows that belongs to different "main_action".
- How many house floor actions are there as part of senate bills (category)?
    - What proportion of these actions were suspended? What proportion of these actions were suspended as amended?
    - The house floor actions mention specific people mentioned by (Mr., Ms., Mrs., etc). Use regular expressions to extract the name of specific people. Produce a new dataset counting how many bills mention the same person and extracting the names of the people mentioned. What is the distribution of total bills proposed per person?
- Now focus a subset of bills whose actions are marked as "senate amendment proposed (on the floor)". An important feature of the text is that it can mention one or more senators.
    - Produce a new dataset counting how many senators are mentioned per bill. This dataset should also include two variables, with the name of the first senator (Senator 1) mentioned in the bill and the name of the second senator (Senator 2). What is the distribution of number of senators per bill?
    - With how many different people does each Senator 1 coauthor bills, i.e. how many different people appear as Senator 2 for each Senator 1?
- Briefly describe your insights.

Think carefully about how to present your results. To facilitate the process of using regular expressions, you may want to use the following additional information on regular expressions:

https://www.dataquest.io/blog/regex-cheatsheet/

https://www.geeksforgeeks.org/python-regex-cheat-sheet/

Deliverable

Submit a Jupyter notebook with you results on Canvas. Include markdown chunks to write comments on your findings and explain your methodology.

When submitting your work imagine that you are sharing this with colleagues at work, who have asked you to explain and justify key decisions and findings of your analysis. The file should be professionally presented, self-explanatory, and have clearly labeled sections.

Rubric:

You will be graded both on the code and the writing using the following point system:

Code - Code runs without issues - Code is appropriate to answer question - File is appropriately organized, with different selections and comments clarifying the definition of variables and dataset names.	4 points
Description of methodology - Explain key coding decisions via markdown chunks - Clarify any choices about how to clean data - Shows evidence of conducting quality checks (count sample size of merged datasets, check for the presence/absence of missing values, and/or compute any relevant descriptive statistics)	2 points
Results and Visualization - Write a short summary of the findings. - Correct interpretation of results and easy to follow - Includes graphs to display results from different groups - All graphs look professional (have axis and title labels, suitable color scheme, font sizes, legend, etc.)	4 points